Ting-Hsuan Chen

Last updated in October 2026

Los Angeles (U.S. Citizen) tchen783@usc.edu (323)791-8418 koi953215.github.io
Google Scholar koi953215 tinghsuan69

Research Interests

Generative AI for embodied intelligence: video diffusion and world models for scaling robot data, physically consistent generation, world-action models that turn prediction into action, and humanoid and multi-agent embodied systems that generalize to unseen situations and coordinate with other agents.

Education

University of Southern California *Aug 2024 – present*
PhD in Computer Science; MS in Computer Science | GPA: **3.92/4.0**

- Advised by Prof. Yue Wang; supported by the **Viterbi School Taiwan Global Fellowship**

National Chung Hsing University *Sept 2019 – June 2023*
Department of Applied Mathematics (DAM) - Computer Science Group

- Final Average **96.6%**, GPA **4.27/4.3**; ranked 1st in all eight semesters; Valedictorian
- Admitted as the representative of the College of Science to the Phi Tau Phi Honor Society

Publications (*: Equal contribution, †: Equal advising)

PwP: Panoramas without Panoramas *Under review*
Camera-Controlled 360° Video from Any View via Perspective-Only Adaptation
Ting-Hsuan Chen[*], Louison Lu[*], Qi Pan[*], Hengyuan Zhang, Xinyu Huang, Cho-Ying Wu, Yue Wang, Liu Ren
Project page

CoLM: Controllable Text-Driven Humanoid Loco-Manipulation Generation *Under review*
Han Jiang[*], Siheng Zhao[*], *Ting-Hsuan Chen*, Yue Wang
Project page

TimePre: Bridging Accuracy, Efficiency, and Stability in Probabilistic Time-Series Forecasting *TMLR 2026*
Lingyu Jiang[*], Lingyu Xu[*], Peiran Li, Dengzhe Hou, Qianwen Ge, Dingyi Zhuang, Shuo Xing, Wenjing Chen, Xiangbo Gao, *Ting-Hsuan Chen*, Xueying Zhan, Xin Zhang, Ziming Zhang, Zhengzhong Tu, Michael Zielewski, Kazunori Yamada, Fangzhou Lin
arXiv

Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion *CVPR 2026*
Ting-Hsuan Chen[*], Ying-Huan Chen[*], Tao Tu, Jie-Ying Lee, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo[†], Yu-Lun Liu[†], Yue Wang[†], Liu Ren
Project page arXiv

Dino-Diffusion Modular Designs Bridge the Cross-Domain Gap in Autonomous Parking *Under review*
Zixuan Wu, Hengyuan Zhang, *Ting-Hsuan Chen*, Yuliang Guo, David Paz, Xinyu Huang, Liu Ren
arXiv Code

MoonSim: A Photorealistic Lunar Environment Simulator *Under review*
Ting-Hsuan Chen[*], Henghui Bao[*], Ziyu Chen[*], Haozhe Lou, Ge Yang, Zhiwen Fan, Marco Pavone, Yue Wang
Project page

NaRCan: Natural Refined Canonical Image with Integration of Diffusion Prior for Video Editing *NeurIPS 2024*

Ting-Hsuan Chen, Jiewen Chan, Hau-Shiang Shiu, Shih Han Yen, Changhan Yeh, Yu-Lun Liu

Project page arXiv Code Demo

DiffIR2VR-Zero: Zero-Shot Video Restoration with Diffusion-based Image Restoration Models *arXiv 2024*

Changhan Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, *Ting-Hsuan Chen*, Yu-Lun Liu

Project page arXiv Code Demo

Experience

Artificial Intelligence Research Intern – Deep Learning *Starting May 2027*

NVIDIA Research

- Incoming research intern working on embodied AI.

Scene Understanding/GenAI – Research Intern *Sunnyvale, United States*

Bosch Center for Artificial Intelligence *May 2025 – Aug 2026*

- Co-first author of **Pantheon360** (CVPR 2026): a 3D-aware 360° video diffusion framework that leverages an explicit point cloud cache to ensure geometric consistency and precise trajectory control for digital twin generation.
- Co-first author of **PwP** (under review): adapts panoramic video generators to new domains using only perspective multi-view videos, generating camera-controlled 360° video from any initial view.
- Engineered a diffusion-based planning system with DINOv2 perception (**Dino-Diffusion**), enabling robust zero-shot generalization in challenging environments.
- Worked on controllable driving behavior simulation that follows goal and sketch prompts without sacrificing safety, and object-centric video generation with accurate physics, fine-grained text control over individual objects, and strong generalization.

PhD Student & MS Student *Los Angeles, United States*

University of Southern California *Aug 2024 – present*

Advised by Prof. Yue Wang. My research turns generative priors into physical intelligence, along three thrusts:

- **Data scaling**: using video diffusion and world models to generate robot data beyond teleoperation.
- **Physical consistency**: pushing generated content to obey real-world physics, not merely look plausible.
- **World-action coupling**: how world models turn prediction into action, via inverse dynamics, joint video-action denoising, or action-only inference with video co-training.

MS Student (Spring 2024)/Research Assistant *Hsinchu, Taiwan*

National Yang Ming Chiao Tung University *Jan 2024 – June 2024*

- Entered Prof. Yu-Lun Liu's laboratory and led a Google-sponsored industry-academia collaboration project.
- Published **NaRCan** as first author at NeurIPS 2024 within five months.
- Collaborated with MediaTek to develop the first zero-shot video restoration framework utilizing diffusion-based image restoration models (**DiffIR2VR-Zero**).

R&D Engineer *Taipei, Taiwan*

Foxconn, Hon Hai Precision Industry *July 2023 – Dec 2023*

- Developed the company's first patented ECG waveform recognition system within three months, integrating AI, computer vision, and signal processing technologies.
- Built an AI-based data cleansing and classification system for medical data, together with its Django APIs.
- Served as a technical mentor for interns and represented the company at medical conferences.

Patents

System and Method of Modular Autonomous Parking Pipeline Using Visual Foundation Models and Diffusion Planners *U.S. Patent Application*

Hengyuan Zhang, Zixuan Wu, *Ting-Hsuan Chen*, Yuliang Guo, David Paz, Xinyu Huang, Liu Ren

Assigned to Robert Bosch GmbH. A modular autonomous parking pipeline that combines visual foundation models with diffusion-based planners (the technology behind Dino-Diffusion).

Data Analysis Method, Apparatus, Electronic Device and Storage Medium *TW113141155, 2024*

Ting-Hsuan Chen

A data analysis method based on feature waveform analysis that improves recognition accuracy through convolution and segmentation operations.

Projects

DreamMesh *GitHub, 2025*

Ting-Hsuan Chen, Cameron Smith, Jiageng Mao, Daniel Wang. A Blender plugin that transforms text or image inputs into complete 3D scenes with rigged objects, AI-generated backgrounds, and automatic object placement.

GPU-SplineTransformer *GitHub, 2022*

A GPU-based SplineTransformer that accelerates large-scale B-spline basis conversion by over 25× compared with CPU-based solutions.

Open-Source Contributions

ViPE *(NVIDIA Toronto AI Lab)* *Aug 2025*

Fixed multi-view SLAM initialization; the patch was merged into the official repository (Issue #11, PR #16).

Academic Service

Conference Reviewer: NeurIPS, ICLR, CVPR *2025 – present*

Honors & Awards

CVPR 2026 Broadening Participation Scholarship	*Apr 2026*
Viterbi School Taiwan Global Fellowship, USC	*Mar 2026*
Viterbi Conference & Research Fund Award, USC	*Dec 2024*
NeurIPS 2024 Scholar Award	*Oct 2024*
Valedictorian, National Chung Hsing University	*June 2023*
Phi Tau Phi Scholastic Honor Society (top graduate; College of Science representative)	*June 2023*
Golden Key International Honour Society	*Apr 2023*
Presidential Award (6 times)	*2020 – 2022*
Dean's List (2 times)	*2020 – 2022*
Ching-O Award	*Nov 2021*
Outstanding Academic Achievement Award	*Nov 2021*
Professor Kuo Jin-Bin Scholarship	*June 2021*
Building Futures Foundation Scholarship	*Oct 2020*
Outstanding Academic Achievement Award	*Oct 2020*
Certificate of Excellent Performance	*July 2020*

Skills

Programming: Python, C, Java, MATLAB, JavaScript, C#

Frameworks & Tools: PyTorch, TensorFlow, CUDA, MuJoCo, Unreal Engine, Blender, Django

Languages: Native in Chinese; Fluent in English